Machine Learning Basics

1. Loss Functions

A loss function measures the gap between a model's predictions and the true targets. Training tries to make the loss function as small as possible. Mean squared error is a common loss function for regression, while cross-entropy is used for classification.

2. Gradient Descent

Gradient descent is an optimization method that updates the weights by moving them against the gradient of the loss. Each step reduces the loss a little, and repeating many steps leads toward a minimum. Gradient descent needs a loss function to tell it which direction is downhill.

3. Learning Rate

The learning rate controls how large each gradient descent step is. A learning rate that is too high makes training overshoot the minimum and diverge, while a learning rate that is too low makes training very slow. Many practitioners decay the learning rate as training progresses.

4. Backpropagation

Backpropagation computes the gradient of the loss with respect to every weight by applying the chain rule layer by layer. Backpropagation reuses intermediate results from the forward pass, which makes it efficient. The gradients from backpropagation are then used by gradient descent to update the weights.

5. Overfitting

Overfitting happens when a model memorizes the training data and fails to generalize to new examples. A widening gap between training loss and validation loss is the classic sign of overfitting. Overfitting is more likely with small datasets and very large models.

6. Regularization

Regularization discourages overfitting by adding a penalty to the loss function. L2 regularization penalizes large weights, which keeps the model smoother. Dropout and early stopping are other forms of regularization.